



Review article

Association between fire smoke fine particulate matter and asthma-related outcomes: Systematic review and meta-analysis[☆]

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ABSTRACT

Background: Asthma-related outcomes are regularly used by studies to investigate the association between human exposure to landscape fire smoke and health. Robust summary effect estimates are required to inform health protection policy for fire smoke exposure.

Objective: To conduct a systematic review and meta-analysis to estimate the association between short-term exposure to landscape fire smoke (LFS) fine particulate matter (PM_{2.5}) and asthma-related outcomes.

Methods: We conducted a systematic review and meta-analysis following PRISMA guidelines. Four databases (PubMed, Medline, EMBASE and Scopus) and reference lists of recent fire smoke and health reviews were searched. The Newcastle-Ottawa Scale was used to evaluate the quality of case-crossover studies, and a previously validated quality assessment framework was used for observational studies lacking control groups. Publication bias was assessed using funnel plots and Egger's Test. The trim and fill method was used when there was evidence of publication bias. Sensitivity and influence analyses were conducted on all endpoints to test the robustness of estimates. Summary estimates were obtained for hospitalisations and emergency department (ED) visits. A descriptive analysis was conducted for physician visits, medication use, and salbutamol dispensations. **Results:** From an initial 181 articles (after duplicate removal), 20 studies were included for quantitative assessment and descriptive synthesis. LFS PM_{2.5} levels were positively associated with asthma hospitalisations (RR = 1.06, 95% CI: 1.02–1.09) and emergency department visits (RR = 1.07, 95% CI: 1.04–1.09). Subgroup analyses found that females were more susceptible than males for ED visits, and that there was an increasing association by age groups for hospital admissions and ED visits. High heterogeneity between studies was observed, but results were robust to sensitivity analysis.

Conclusions: Females and all adults aged over 65 years appear to be the population groups most sensitive to asthma-related outcomes when exposed to LFS PM_{2.5}. Overall, results were higher than those obtained for a typical PM_{2.5} mixture.

1. Introduction

Asthma is a diverse condition which generally involves chronic inflammation of the airways. It is defined by the Global Initiative for Asthma (GINA) (2018 p. 14) as “the history of respiratory symptoms such as wheeze, shortness of breath, chest tightness and cough that vary

over time and in intensity, together with variable expiratory airflow limitation”. In 2016, it was estimated that the prevalence of asthma was more than 330 million people (GBD 2016 Disease and Injury Incidence and Prevalence Collaborators, 2017), and that this translated into more than 23 million disability adjusted life years (DALYs), positioning asthma in the 28th position among the leading causes of burden of

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disease (Global Asthma Network, 2018; Global Initiative for Asthma, 2018). Evidence shows that prevalence is higher in males during childhood but increases in females after puberty (Ferrante and La Grutta, 2018). In the case of older people, evidence is scarce, but shows that asthma is usually more severe and associated with other comorbidities such as chronic obstructive pulmonary disease (COPD) (Battaglia et al., 2016; Curto et al., 2019). Healthcare systems and society are stressed through increased hospitalisations, emergency department visits, medication dispensations, morbidity and mortality. Families and the workplace could be potentially economically affected due to loss of productivity (Global Initiative for Asthma, 2018). Ultimately, these impacts translate to high average per patient yearly costs (Nunes et al., 2017), such as the estimated EUR 1583 (in 2010 euros) for adults aged 30–54 years in Europe (Accordini et al., 2013) or the USD\$ 3266 (in 2015 U.S. dollars) for the general population in the USA (Nurmagambetov et al., 2018).

Indoor and outdoor air pollution has been recognised as an important environmental risk factor associated with asthma (Beasley et al., 2015). It is estimated that globally between 5 and 10 million asthma emergency department visits could be attributable to fine particulate matter (PM_{2.5}), representing 4–9% of the annual number of global visits (Anenberg et al., 2018). PM_{2.5} from landscape fire smoke (LFS) has been associated with a myriad of health effects, diverse in type and magnitude, ranging from all-cause mortality to respiratory hospital admissions and emergency department (ED) visits and increased use of medication for treating asthma and asthma-like symptoms (Adetona et al., 2016; Black et al., 2017; Cascio, 2018; Liu et al., 2015; Reid et al., 2016a). Asthma-related outcomes are amongst the most reported endpoints.

Health impact assessments, cost-effectiveness analyses, and cost-benefit analyses are commonly used methodologies for estimating burden of disease and health costs associated with air pollution and for analysing different pollution (exposure) reduction strategies. They are particularly useful when informing environmental and public health policies (Cárdaba Arranz et al. 2014; Martuzzi et al., 2003). Currently, there are no recommended concentration response functions (CRFs) for quantifying asthma-related outcomes attributable to particulate matter (PM) for conducting health impact assessments, except for the incidence of asthma symptoms in asthmatic children and daily mean PM₁₀ (World Health Organization Regional Office for Europe, 2013). Furthermore, LFS PM_{2.5}, which is composed of multiple chemicals produced during biomass combustion (Johnston et al., 2012), is different to a typical or multisource PM_{2.5} mixture which includes different chemicals produced by the combustion of multiple sources and types of fuels. Recent systematic reviews and meta-analyses have calculated pooled effect estimates for the short-term effect of PM_{2.5} on asthma-related hospital admissions and ED visits, identifying positive and stronger associations for males, children and elders (Fan et al., 2016; Lim et al., 2016; Zheng et al., 2015). Nevertheless, these studies were not focused on LFS or a smoke-dominated PM_{2.5} mixture, but rather on general multi-source air pollution. Previous studies have found that asthma-effects related to LFS PM are higher than non-LFS PM (Alman et al., 2016; DeFlorio-Barker et al., 2019; Gan et al., 2017; Johnston et al., 2014; Morgan et al., 2010; Reid et al., 2016b).

With an increasing expectation of more intense and frequent fires in the future, and an increasing use of prescribed burns for the reduction of wildfire risk, there is a constant need for evidence that will adequately help inform policy. The purpose of this study was to conduct a systematic review and meta-analysis of the associations between short-term exposure to LFS PM_{2.5} and asthma-related outcomes, including hospital admissions, emergency department visits, physician visits, ambulance dispatches, medication use, and salbutamol dispensations. When possible, we performed subgroup analyses to obtain summary estimates by age group, sex, lags, country, study design, and exposure method.

2. Methods

The PRISMA protocols for systematic reviews and meta-analysis (Moher et al., 2015) were used, and the protocol was registered in PROSPERO¹ under registration ID CRD42018108767 at the beginning of the study.

2.1. Study question

We aimed to answer the following question: “What is the impact of short-term exposure to LFS PM_{2.5} on asthma-related outcomes such as hospital admissions, emergency department visits, physician visits, ambulance dispatches, medication use and salbutamol dispensations”. The detailed PECOS statement (Morgan et al., 2018) is presented in Table 1.

2.2. Data sources and search strategy

We searched articles published in PubMed, Medline (OVID), EMBASE and Scopus, indexed online to 4th of September of 2018 with no specified start date. The databases were searched again on the 2nd of May of 2019 and no relevant additional publications were identified. The search strategy focused on finding studies that had estimated coefficients linking LFS particulate matter with the following asthma-specific health outcomes: hospital admissions, ED visits, medication use, salbutamol dispensations, ambulance dispatches and physician visits. A combination of the following terms were used to search keywords, title and abstract: “asthma”, “wheez”, “fire”, “wildfire”, “bushfire”, “contamin”, “pollut”, “PM”, “PM10”, “PM2.5”, “particulate matter”, “hospital”, “hospitali”, “admission”, “emergenc”, “physicia”, “dispensatio”, “visit”, “salbutamol”, “attendanc”, “ambulanc”. Full detailed strategies for each database are shown in the supplemental material (Table S2). Additionally, previous systematic reviews done by Liu et al. (2015), Adetona et al. (2016), Reid et al. (2016a), Black et al. (2017) and Cascio (2018) were revised to identify additional studies containing asthma-related coefficient estimations.

2.3. Study selection

Screening was done independently by two researchers (NB and JH) in a two-stage process. First, a title and abstract screening was done, classifying each study with a ‘yes’, ‘no’ or ‘maybe’. Secondly, a full text screening was done on all studies classified as ‘yes’ or ‘maybe’. Disagreements were discussed with and resolved by a third researcher (FJ). Inclusion criteria consisted of: (1) particulate matter was produced by LFS from some sort of biomass (wildland, forest, sugar cane, peat, etc.); (2) studies presented estimates for asthma-related hospital admissions, ED visits, medication use, salbutamol dispensations, ambulance dispatches or physician visits; (3) for all, except salbutamol dispensations and medication use, asthma cases were identified using the Primary International Classification of Diseases versions 9 or 10 (ICD9 or ICD10), with the following endpoints: asthma (ICD9: 493 or ICD10: J45-J46); (4) exposure metric consisted on PM₁₀ or PM_{2.5} daily average, and (5) study was published in a scientific journal. Studies were excluded if they were not full text scientific articles, presented no effect estimates, were not epidemiological studies, did not deal with vegetation or landscape fires, or were reassessments of original and previous studies. We only included studies that presented estimates as % increase, relative risk or risk ratio (RR) and odds ratio (OR). We did not exclude studies by design. Screening was done using the Covidence software (Veritas Health Innovation, 2019), and references for full text screening were managed using Mendeley Desktop.

¹ <http://www.crd.york.ac.uk/PROSPERO/>.

Table 1
PECOS statement.

PECOS statement:	<u>Population:</u> all human population <u>Exposure:</u> exposure to LFS PM_{2.5}. <u>Comparator(s):</u> Reference groups exposed to lower or no levels of LFS PM_{2.5}, or same-subject exposure during times where the health outcome of interest did not occur (e.g. in case-crossover study designs) <u>Outcome:</u> Asthma-related hospital admissions, emergency department visits, physician visits, ambulance dispatches, medication use and salbutamol dispensations <u>Study Design:</u> Any study design
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2.4. Data extraction

Data extraction used two predefined templates produced by the authors. The first, consisted on general study characteristics, and the following data fields were included: (1) year of publication, (2) first author, (3) geographic scope, (4) study period, (5) study design, (6) statistical model, (7) confounders considered, (8) health outcomes. The second focused on results and the following data fields were included: (1) health outcome, (2) exposure metric, (3) exposure method, (4) age group, (5) sex, (6) indigenous status, (7) air quality range, (8) pollutant, (9) measure, (10) lag, (11) case count, (12) estimate, (13) upper and lower 95% confidence intervals, (14) delta PM concentration, (15) p-value, and (16) PM_{2.5}/PM₁₀ ratio. When studies presented results for multiple cities, estimates were extracted for each city. When detailed estimates were presented only in figures ($n = 4$), corresponding authors were contacted to obtain results (Gan et al., 2017; Haikerwal et al., 2016; Hanigan et al., 2008; Henderson et al., 2011). When authors were unable to provide detailed coefficient estimates (Haikerwal et al., 2016; Henderson et al., 2011), these were extracted from 3s using the Web-PlotDigitizer software (Moeyaert et al., 2016; Rohatgi, 2010).

2.5. Quality assessment and risk of bias

Quality of studies was assessed using the Newcastle-Ottawa Quality Assessment Scale (NOS) in the case-crossover studies. The NOS is used to assess eight specific items grouped into three more general quality parameters (selection, comparability, and outcome). Usually each individual item is graded with a maximum of one point, but comparability could score a maximum of two points. When studies score less than 5 out of 9 points, it is said that there is a high risk of bias (Luchini et al., 2017). For other observational designs (including time series, fire episode studies and panel studies), a modified version of a previously validated assessment framework was applied. The framework developed by Zaza et al. (2000), assesses study design, validity and reliability, generalisability, risk of bias and reporting, and has been previously used to assess quality of observational studies that lacked control groups in systematic reviews (Bowatte et al., 2018; Lambert et al., 2017; Tham et al., 2014). This latter framework is more detailed compared to the NOS, and was therefore also applied to case-crossover studies, to see if we could identify any additional elements that could have influenced the quality of these studies, and to allow comparability between case-crossover and other observational designs. Quality assessment was carried out independently by two authors (NB and RT).

Publication bias was assessed using funnel plots and Egger's Test (Egger et al., 1997). Additionally, the "trim and fill" method (Duval and Tweedie, 2000) was used to explore the impact of adjusting for potential publication bias.

2.6. Data synthesis

A quantitative data synthesis was performed only when three or more studies were available for each health endpoint, a descriptive

analysis was done in other cases. Relative risk (RR) and the 95% confidence intervals (CI) per $10 \mu\text{g}/\text{m}^3$ of PM_{2.5} were derived from all studies and used to estimated summary measures weighted by the inverse of the variance, giving less weighting to studies with high variance. When original results were presented by an increase of PM₁₀, the respective increase in terms of PM_{2.5} was calculated using the PM_{2.5}/PM₁₀ ratio. This ratio was obtained from the original study, when possible, or from other studies conducted in same geographic area. There was no single lag used in all studies, lag 0 was the most commonly reported. If only one lagged estimate for a given pollutant/outcome pair was reported, this estimate was considered for the overall analysis. If more than one lag measure was presented, we selected one for the main meta-analysis according to the following priority: (1) lag 0, (2) the lag that the author focused on or stated a priori. To account and allow for heterogeneity between studies a DerSimonian and Laird random effects model was used (Viechtbauer, 2010).

The main analysis consisted on obtaining a pooled RR estimate per $10 \mu\text{g}/\text{m}^3$ of PM_{2.5} for all sexes and all ages. Subgroup analyses were done by age groups, sex, lags, country, study design and exposure method. The age groups considered were children (0–18 years), adults (19–64 years), and elders (65 + years). Age grouping in the studies were assigned to one of these three age groups whenever possible (e.g. 0–14 years to children, 15–64 years to adults, 66 + years to elders). Different exposure methods were identified, including the use of WRF chemical models, CALPUFF models, surface monitoring with kriging, and surface monitoring with city-wide averages. We classified each method as modelled or monitors.

To test robustness of estimates a sensitivity and influence analysis was done. The sensitivity analysis consisted on excluding one study at a time and re-computing summary estimates. The influence analysis consisted on the estimation and visual analysis of different diagnostics to identify studies that strongly influenced the overall results (Harrer and Ebert, 2018; Viechtbauer and Cheung, 2010). All analyses were done using the 'metafor' package (version 2.0–0) and the R software version 3.5.1 (R Core Team, 2018).

3. Results

3.1. Study characteristics

The search identified a total of 181 articles after duplicates were removed. After performing the title and abstract screening, 60 studies were assessed for full-text eligibility, out of which only 20 were included for the quantitative and descriptive synthesis (see Fig. 1).

Table 2 presents a summary of characteristics of studies included in the quantitative and descriptive analyses. Most studies ($n = 15$) estimated effects for hospital admissions, ED visits or both, and were done either in Australia ($n = 7$) or the USA ($n = 8$). Five studies estimated effects for medication use, salbutamol dispensations and/or physician visits, and were conducted in Canada ($n = 4$) or Australia ($n = 1$). Of the 20 included studies, 60% ($n = 12$) were published during or after 2013, while one was published in 2002 (Johnston et al., 2002) and the most recent one in 2018 (Hutchinson et al., 2018). No studies were found regarding asthma-related ambulance dispatches.

The quality assessment scores are presented in Tables S3, S4, and S5 in the supplemental material. Case-crossover studies obtained excellent results when applying the NOS (Table S3), whilst other observational design studies obtained average to high scores using the modified version of the framework developed by Zaza et al. (2000) (Table S4). One study stood out for having a lower score (Resnick et al., 2015), with 11 out of 24 points, particularly for not incorporating multilevel analyses and limited accounting for covariates in the models. Delfino et al. (2009) obtained the highest score amongst other observational design studies, with 19 out of 24 points. When assessing case-crossover studies with this same tool (Table S5) all scores were either 21 or 22, being considerably higher than other observational design studies. In

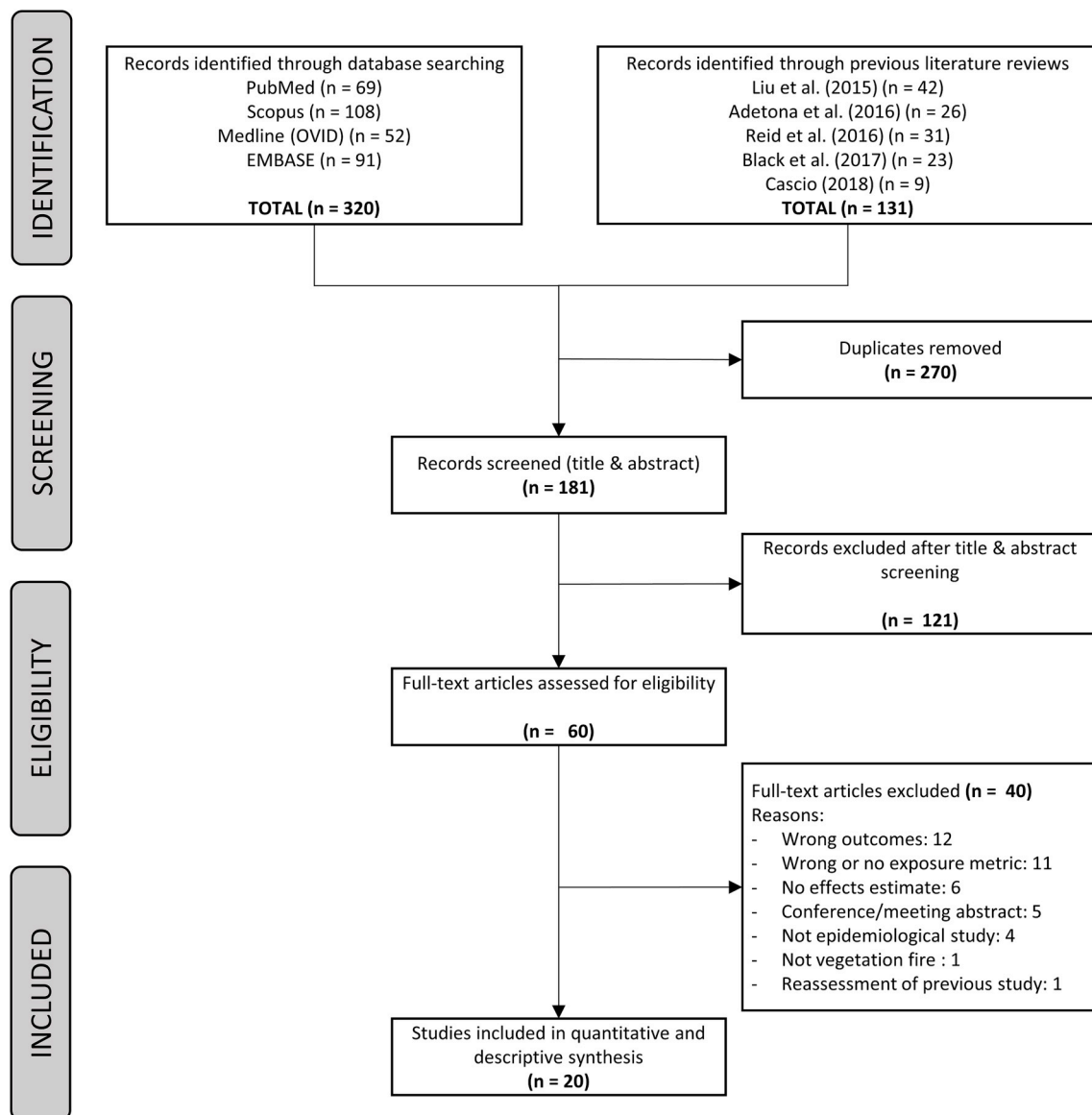


Fig. 1. PRISMA flowchart.

general, quality of studies was good or very good, and main differences were related to the way confounding and biases, particularly exposure misclassification, were managed and discussed.

For hospital admissions, all studies presented a general estimate for the broader population or stratified by all age groups, except one (Le et al., 2014), which only estimated a relationship for elders (65 + years). Only two studies showed results stratified by sex (Delfino et al., 2009; Reid et al., 2016b), so no analysis was done for these subgroups. One study (Hanigan et al., 2008) presented estimates for indigenous and non-indigenous Australians, but not for the overall population. Five studies used other observational designs (Delfino et al., 2009; Hanigan et al., 2008; Le et al., 2014; Morgan et al., 2010; Reid et al., 2016b), while two used case-crossover designs (Johnston et al., 2007; Martin et al., 2013). All studies adjusted for temperature, four studies for influenza, three studies for public holidays, one study adjusted for fungal spores (Delfino et al., 2009), and no studies adjusted for pollen. The hospital admissions meta-analysis included summary estimates for: (1) all ages and all sexes; (2) age groups (children, adults, elders); (3) single-day lags (0,1,2,3), (4) country (Australia or USA), (5) study design (case-crossover or other observational designs), and (6) exposure method (modelled or monitors).

For ED visits, all studies estimated coefficients for the general population and by age groups covering the whole population, except one study that showed results for the general population not stratified by age group (Johnston et al., 2002). Three studies presented estimates stratified by sex (Gan et al., 2017; Haikerwal et al., 2016; Reid et al., 2016b). Studies were similarly distributed by study design between time series (n = 4) and case-crossover (n = 5). Most studies of ED visits (n = 6) adjusted for temperature, two studies adjusted for holidays (Johnston et al., 2014; Reid et al., 2016b), one study only adjusted for weekly rates of influenza and for weekend/weekday (Johnston et al., 2002), and no studies adjusted for fungal spores or pollen. The ED visits meta-analysis included summary estimates for: (1) all ages and all sexes; (2) by age groups (children, adults, elders); (3) by sex (males, females); (4) by single-day lags (0,1,2), (5) by country (Australia or USA), (6) by study design (case-crossover or other observational designs), and (7) exposure method (modelled or monitors).

For physician visits, three studies (1 cohort and 2 time series) considered different fire periods in British Columbia, Canada (Henderson et al., 2011; Yao et al., 2013, 2016), and one study analysed a fire period in San Diego, California (Hutchinson et al., 2018). Canadian studies adjusted for temperature and day of week, while

Table 2
Study characteristics.

Reference	Geographic scope	Study period	Season	Exposure method	Study design	Statistical Model	Confounders	Health Outcome(s)	Diagnosis Group (s)
Alman et al. (2016)	Colorado, USA	June 5 to July 6, 2012	Summer	WRF-chemical model	Case-crossover	Conditional logistic regression	Temperature, day of week	ED visits	Asthma and wheeze (ICD9: 493-786.07)
Delfino et al. (2009)	Southern California, USA	October 1 to November 15, 2003 (wildfire period: 21–30 October)	Autumn	Surface monitors (with spatial interpolation)	Time series	Generalised estimating equations for Poisson data	Relative humidity, temperature and surface pressure gradient. Fungal spores. Temporal trends	Hospital admissions	Asthma (ICD9: 493)
Elliott et al. (2013)	British Columbia, Canada	Fire season: 2003–2010	Spring-Summer	Surface monitors	Time series	Generalised linear models	Temperature, relative humidity, and temporal trends.	Salbutamol dispensations	
Gan et al. (2017)	Washington, USA	1 July to 31 October 2012	Summer-Autumn	3 methods: GWR smoke, Surface monitors (with Kriging), WRF-chemical model	Time -stratified case-crossover	Conditional logistic regression	Temperature, relative humidity, wind speed, and precipitation	ED visits	Asthma (ICD9: 493)
Haikerwal et al. (2016)	Victoria, Australia	2006–2007 wildfire period	Summer	atmospheric dispersion air pollution model (TAPM) coupled with a chemical transport model (CTM)	Time -stratified case-crossover	Conditional logistic regression models	Temperature, relative humidity	ED visits	Asthma (ICD10: J45-J46)
Hanigan et al. (2008)	Northern Territory, Australia	1996–2005 fire season	Autumn-Winter-Spring	Surface monitors	Time series	Over-dispersed Poisson generalised linear model	Daily temperature, relative humidity, day of week, influenza epidemic	Hospital admissions	Asthma (ICD9: 493, ICD10: J45-J46)
Henderson et al. (2011)	British Columbia, Canada	1 July to 30 September 2003 (2003 fires)	Summer	3 methods: Calpuff, presence of smoke, surface monitors	Time series	Logistic regression	Same-day mean temperature, day of week, and week of study	Physician visits	Asthma (ICD9: 493)
Hutchinson et al. (2018)	California, USA	August 16 and December 15, 2007 (fire period late October 2007)	Summer-Autumn	coupled wildfire smoke emissions and atmospheric dispersion models	Case-crossover	Conditional logistic regression	Daily temperature and relative humidity	ED visits Hospital admissions	Asthma (ICD9: 493)
Johnston et al. (2002)	Northern Territory, Australia	1 April – 31 October 2000	Autumn-Winter-Spring	Surface monitors	Ecological	Negative binomial regression	weekly rates of influenza and for weekend or weekday	Hospital admissions	Asthma (ICD9: 493.00 and 493.9)
Johnston et al. (2006)	Northern Territory, Australia	Seven-month period 2004	Autumn-Winter-Spring	Surface monitors	Panel study	Logistic regression (OR for dichotomous outcomes) Negative binomial regression (RR for count outcomes)	minimum daily air temperature, relative humidity, pollen and spore counts, the weekly rate of consultations to general practitioners for influenza-like illness, temporal autocorrelation of outcomes, weekends and holiday periods	Medication use	
Johnston et al. (2007)	Northern Territory, Australia	Three fire seasons of 2000, 2004 and 2005	Autumn-Winter-Spring	coupled wildfire smoke emissions and atmospheric dispersion models	Time -stratified case-crossover	Conditional logistic regression	weekly influenza rate, days with rainfall > 5 mm, same day mean temperature and humidity, the mean temperature and humidity of the previous three days and public holidays	Hospital admissions	Asthma (ICD10: J45-J46)
Johnston et al. (2014)	New South Wales, Australia	1 July 1996 to 30 June 2007 (46 smoke event days)	All year	Surface monitors (city-wide average)	Time -stratified case-crossover	Conditional logistic regression	temperature and dew point (both same-day and the average for the previous three days), influenza epidemics, public holidays, and school holidays	ED visits	Asthma (ICD9: 493, ICD10: J45-J46)
Le et al. (2014)	81 counties in 11 North-Eastern and Mid-Atlantic States, USA	1/07/2002	Summer	Surface monitors (with Kriging)	Ecological (fire episode)	Poisson regression	PM2.5, temperature, and dew point	Hospital admissions	Asthma (ICD9: 493)
Martin et al. (2013)	New South Wales, Australia	1994–2007	Spring-Summer-Autumn	Surface monitors (city-wide average)	Time -stratified case-crossover	Conditional logistic regression	Temperature, dew point, influenza epidemics, public holidays, school holidays	Hospital admissions	Asthma (ICD9: 493, ICD10: J45-J46)
		1994 through 2002			Time series			Hospital admissions	

(continued on next page)

Table 2 (continued)

Reference	Geographic scope	Study period	Season	Exposure method	Study design	Statistical Model	Confounders	Health Outcome(s)	Diagnosis Group (s)
Morgan et al. (2010)	New South Wales, Australia	June 1 - July 14, 2008	Summer-Autumn	Surface monitors (city-wide average)	Ecological (fire episode)	Poisson model, with bootstrap based methodology	Temperature, humidity, day of week, flu epidemic	Asthma (ICD9: 493, ICD10: J45-J46)	
Rappold et al. (2012)	North Carolina, USA	May 6 - September 15, 2008	Summer	Smoke dispersion model	Ecological (fire episode)	Poisson mixed effects regression	None listed	ED visits	Asthma (ICD9: 493)
Reid et al. (2016b)	California, USA	reference: May 1 - May 31. Fire period: June 1 - June 13	Summer	Exposure prediction model	Ecological (fire episode)	Poisson generalised estimation equations	Temperature, RH, Smoking prevalence, Daily 8-hr max Ozone, Holiday, Day of Week, Median Income (SES), percent of population > 65, Heat Index.	Hospital admissions, ED visits	Asthma (ICD9: 493)
Resnick et al. (2015)	New Mexico, USA	reference: May 1 - May 31. Fire period: June 1 - June 13	Summer	Surface monitors	Ecological (fire episode)	Poisson regression	None listed	ED visits	Asthma (ICD9: 493)
Yao et al. (2013)	British Columbia, Canada	24 July - 29 August 2010	Summer	3 methods: Bluesky forecasting model, Surface monitors, truncated bluesky forecasting	Time series	Poisson regression	Same-day maximum temperature, day-of-week, holiday, and week-of-study	Salbutamol dispensations, Physician visits	
Yao et al. (2016)	British Columbia, Canada	forest fire seasons (1 April to 30 September) of 2003 through 2010	Spring-Summer	2 methods: modelled, surface monitors	Time series	Poisson regression	Temperature, year, month, day of week	Salbutamol dispensations, Physician visits	

Hutchinson et al. (2018) adjusted for daily temperature and relative humidity. One study analysed the summer of 2003 (Henderson et al., 2011), one study analysed the summer of 2010 (Yao et al., 2013), one study analysed all fire seasons from 2003 to 2010 (Yao et al., 2016) including the time frames of the previous two studies, and one study analysed the 2007 fire season (Hutchinson et al., 2018). While Henderson et al. (2011) and Hutchinson et al. (Hutchinson et al., 2018) estimated effects on different age groups, the other two studies focused on analysing different methods and levels of exposure.

For salbutamol dispensations, three time series studies focused on fires in British Columbia, Canada at different time frames, with two studies considering all fire seasons between 2003 and 2010 (Elliott et al., 2013; Yao et al., 2016), including the summer of 2010 that was analysed in the other study (Yao et al., 2013). The three studies presented aggregated results for all ages and all sexes. One panel study presented estimates for medication use in the Northern Territory, Australia, during the 2004 fire season (Johnston et al., 2006). Results were presented for all ages, children (0–18) and adults (18+) and included the following outcome measures: proportion of group that (1) used reliever medication, (2) commenced a reliever after at least 7 days without any reliever, (3) commenced a course of oral steroids; and mean number of times a (4) reliever medication was used. No meta-analysis was performed for physician visits, medication use or salbutamol dispensations.

3.2. Relative risks for hospital admissions

Six studies conducted in eight different locations were included for this meta-analysis. Two studies (Hanigan et al., 2008; Johnston et al., 2007) analysed the association between LFS PM₁₀ and hospital admissions in Darwin, Australia. Johnston et al. (2007) was selected over Hanigan et al. (2008) for the overall analysis, but the latter was considered for the analysis by single-day lags. Results at four locations reported positive associations.

Summary results for the meta-analysis with 95% confidence intervals (95% CI) for an increase of 10 µg/m³ of LFS PM_{2.5} are presented in Fig. 2. Detailed forest plots, funnel plots and sensitivity analysis are presented in the supplemental material (see Figs. S1–S14). Sensitivity analysis for all ages and sexes showed that there is a very likely positive association between LFS PM_{2.5} and asthma-related hospital admissions. This was true even when Reid et al. (2016b), which was the most influential study, was removed (see Fig. S30 for influence analysis plots for hospital admissions).

The random effects summary RR for all ages and all sexes was 1.06 (95% CI, 1.02–1.09, p-value (I^2) = 0.01, I^2 = 60.9%, p-value (Egger's) = 0.498), with the Reid et al. (2016b) having a strong influence. Removing this study resulted in an RR of 1.04 (95% CI, 1.02–1.05, p-value (I^2) = 0.67, I^2 = 0%, p-value (Egger's) = 0.574), and results were still positive and statistically significant. By age groups, the effects were positive in elders (RR = 1.12, 95% CI: 1.06–1.17) and adults (RR = 1.08, 95% CI: 1.03–1.14), but not children (RR = 1.03, 95% CI: 0.98–1.08), although there seems to be a tendency towards a positive association in the latter. These positive associations were confirmed by the sensitivity analysis. For lag 0, compared to other single-day lags, the mean effect size was larger but had a wider confidence interval (RR = 1.07, 95% CI: 1.01–1.13), and there seemed to be a positive effect up to one (RR = 1.03, 95% CI: 1.01–1.05), two (RR = 1.03, 95% CI: 1.01–1.05), and three days after exposure (RR = 1.02, 95% CI: 1.00–1.04). Interestingly, summary estimates were larger but more uncertain for studies analysing fires and impacts in the USA, compared to Australia, but in both cases, they remain positive. We observed a similar behaviour between other observational designs and case-crossover study designs, which is coherent with the fact that the two case-crossover studies were done in Australia and one of them included a large population (Sydney) and long period of analysis (1994–2007) (Johnston et al., 2007; Martin et al., 2013). Results by

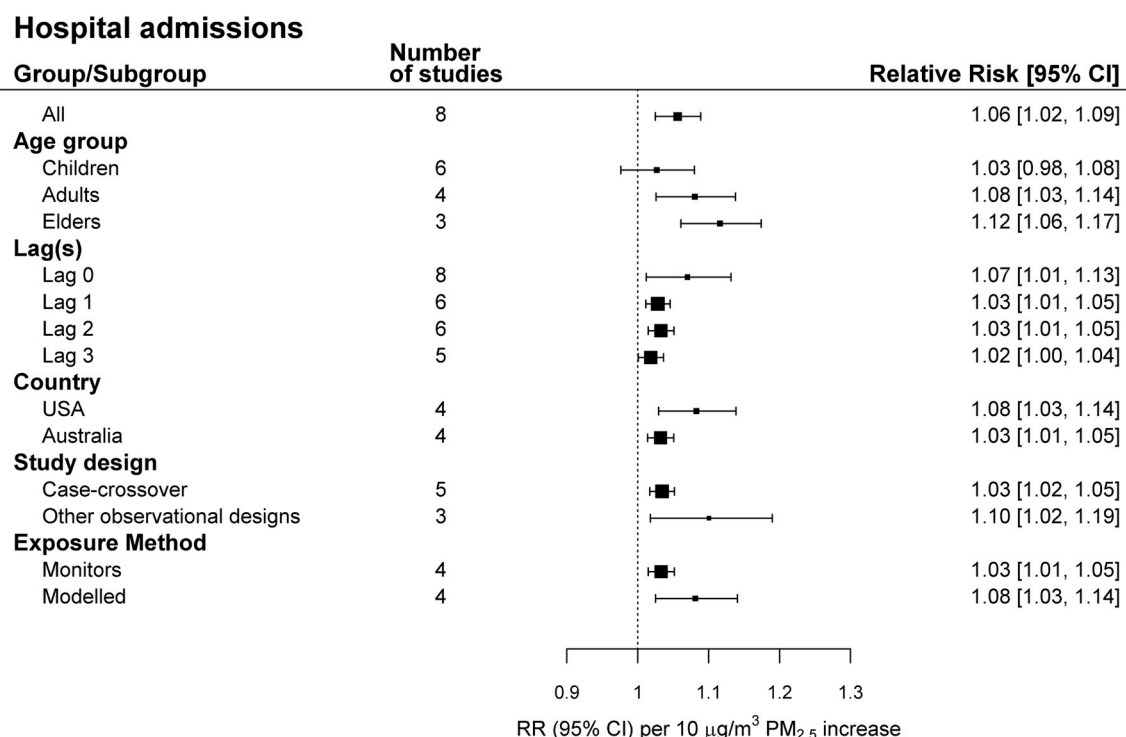


Fig. 2. Summary forest plot for association of asthma hospital admissions with LFS $\text{PM}_{2.5}$. Random effects estimate (95% CI) per 10 $\mu\text{g}/\text{m}^3$ increase in LFS $\text{PM}_{2.5}$.

exposure method varied considerably whether this was determined by monitors (RR = 1.03, 95% CI: 1.01–1.05) or modelled (RR = 1.08, 95% CI: 1.03–1.14), with a lower and more precise association in the case of monitors.

3.3. Relative risks for emergency department visits

Eight studies were included for the ED visits meta-analysis, and summary results are presented in Fig. 3. Detailed forest plots, funnel plots and sensitivity analysis are presented in the supplemental material (see Fig. S15–S29). Sensitivity analysis showed that results were consistently positive and higher than for hospital admissions.

The random effects summary RR for all ages and all sexes was 1.07 (95% CI, 1.04–1.09, p-value (I^2) = 0, I^2 = 82.4%, p-value (Egger's) = 0.049). Publication bias was not detected when applying the trim and fill method. Reid et al. (2016b) was the most influential study, but when this was removed results only had a slight variation to an RR of 1.06 (95% CI, 1.03–1.08) (see Figs. S15 and S31 for detail on sensitivity and influence analysis).

Results showed a positive and increasing effect for children (RR = 1.04, 95% CI: 1.00–1.08), adults (RR = 1.07, 95% CI: 1.04–1.11) and elders (RR = 1.15, 95% CI: 1.10–1.20). Unlike hospital admissions, effects seemed to happen on the same day as exposure (RR = 1.07, 95% CI: 1.04–1.11) but not on the following days. Results for the USA and Australia were similar in magnitude but had a wider confidence interval for Australia. Consistently, results were positive in both cases. With respect to study designs, case-crossover studies showed lower but more certain results (RR = 1.05, 95% CI: 1.03–1.08), compared to other observational designs (RR = 1.12, 95% CI: 1.02–1.23). This could be explained by the fact that case-crossover studies for ED visits included large populated areas in California, New Mexico, Washington, New South Wales and Victoria. Different exposure methods produced different pooled RRs, with a high level of overlap, with an RR of 1.09 (95% CI, 1.03–1.15) for monitors and 1.06 (95% CI, 1.03–1.10) for modelled. This time summary estimate with modelled exposure method resulted in a slightly lower RR and a more precise summary estimate, although there is a high level of overlap.

3.4. Physician visits

Out of the four studies, all of them presented statistically significant associations between LFS PM and asthma-related physician visits. To compare studies, original OR and RRs were normalized to RR for an increase in 10 $\mu\text{g}/\text{m}^3$ of $\text{PM}_{2.5}$. Detail of re-estimated RR per 10 ($\mu\text{g}/\text{m}^3$) increase in LFS $\text{PM}_{2.5}$ and asthma-related physician visits are presented in the supplemental material (see Fig. S32). Three studies analysed how the use of different exposure methods affected the summary estimates (Henderson et al., 2011; Yao et al., 2013, 2016). Consistently, results when exposure was obtained from surface monitors were higher and less precise than those obtained when exposure was modelled. RRs for all ages and all sexes showed a positive association and ranged from 1.01 (95% CI, 1.01–1.02) to 1.11 (95% CI, 1.06–1.15). For age groups, results were positive, except for four age groups (< 5 years, 10–19 years, 50–59 years, and 70–79 years) in the case of Henderson et al. (2011) and one age group (18–64 years) in the case of Hutchinson et al. (2018).

3.5. Salbutamol dispensations and medication use

All results presented by the three studies (Elliott et al., 2013; Yao et al., 2013, 2016) demonstrated a positive and statistically significant association between LFS $\text{PM}_{2.5}$ and salbutamol dispensations, independent of the exposure method used (see Fig. S33). Yao et al. (2016) evaluated the longest period (2003–2010) and estimated that a 10 mg/m^3 increase in LFS $\text{PM}_{2.5}$ was positively associated with salbutamol dispensations with an RR of 1.04 (95% CI, 1.03–1.06). Elliott et al. (2013) found that effect estimates were higher during extreme fire events (RR = 1.07, 95% CI: 1.05–1.1) compared to the fire season average (RR = 1.06, 95% CI: 1.05–1.08).

The only study that assessed association between LFS $\text{PM}_{2.5}$ and medication use (Johnston et al., 2006) found no associations between LFS $\text{PM}_{2.5}$ and the use of reliever medication or the mean number of times reliever medication was used each day, but a positive association was reported for commencing the use of reliever medication and oral steroids (after at least one week without using reliever or oral steroids),

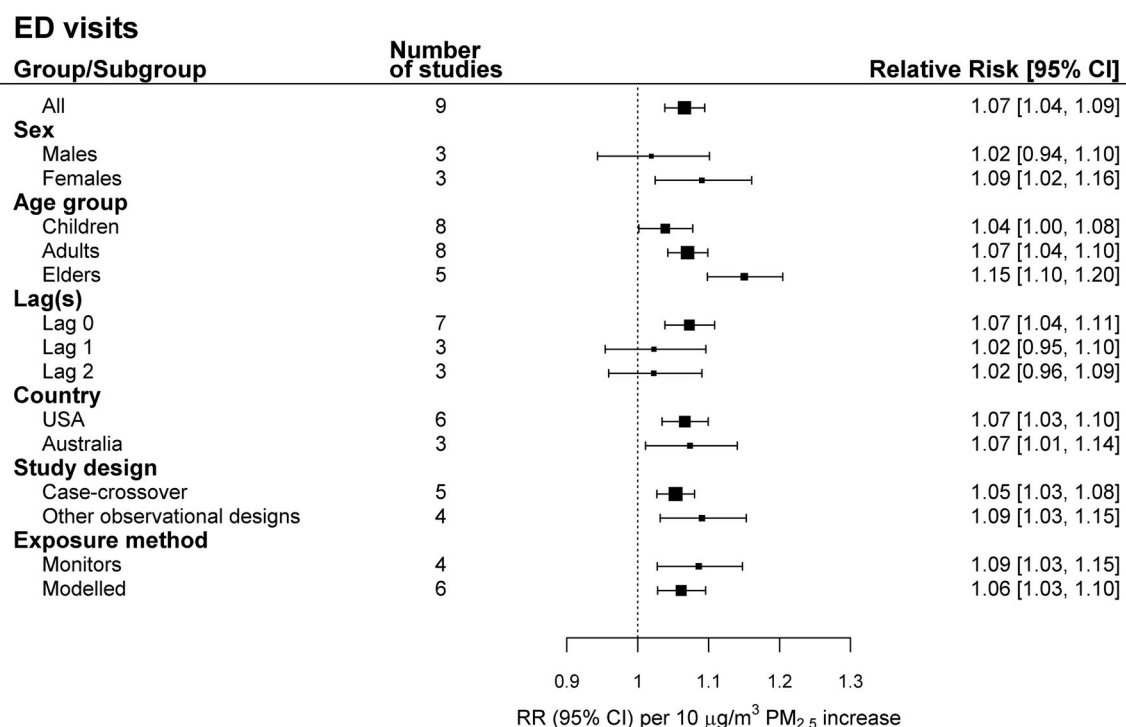


Fig. 3. Summary forest plot for association of asthma ED visits with LFS PM_{2.5}. Random effects estimate (95% CI) per 10 µg/m³ increase in LFS PM_{2.5}.

particularly for adults. (see Fig. S34).

4. Discussion

Our results suggest that short term exposure to LFS PM_{2.5} is positively associated with asthma-related outcomes, and that this association varies by sex, age group, country of study, and study design. A positive association was found for hospital admissions in all sexes and all ages, adults, and elders. Point estimates were higher for same day (lag 0) compared to 1, 2, and 3 days after exposure, but positive in all situations. Results varied between countries (Australia and USA) and study design (case-crossover and other observational designs), but were positive in all cases. For ED visits, a positive association was found for all sexes and all ages, females, all age groups in an increasing magnitude, and for same day (lag 0), but not for 1 and 2 days after exposure. Results remained positive in subgroup analyses by country and study design. Although evidence is limited, our descriptive analysis suggests there is a likely positive association between LFS PM_{2.5} with physician visits and salbutamol dispensations, and for medication use in adults.

For hospital admissions and ED visits, we found a stronger association for elders compared to adults, and for adults compared to children. Differences by age group are particularly relevant for ED visits, with elders having an RR of 1.15 (95% CI, 1.1–1.2) per 10 µg/m³ increase in PM_{2.5}, compared to 1.04 (95% CI, 1–1.08) for children. These results were different in magnitude and direction compared to evidence related to multisource air pollution where the elderly population (RR = 1.022, 95% CI: 1.014–1.031) had a smaller effect compared to children (RR = 1.025, 95% CI: 1.013–1.037) and adults (RR = 1.027, 95% CI: 1.007–1.047) for hospital admissions and ED visits combined (Zheng et al., 2015), and children (RR = 1.036, 95% CI: 1.018–1.053) being more susceptible than adults (RR = 1.017, 95% CI: 1.007–1.028) for ED visits alone (Fan et al., 2016). In ED visits, our results suggest that females are considerably more susceptible than males, whilst the combined effect for hospital admissions and ED visits from multisource pollution produced similar effect estimates for females and males (Zheng et al., 2015). We found higher effects in the USA compared to Australia for hospital admissions but similar in the

case of ED visits. We obtained different associations, although all positive, depending on exposure method used, for both hospital admissions and ED visits. Results were practically identical to those obtained by country. These differences could be explained by differences in the types of combustible biomass, the healthcare systems, and the different levels of exposures in both countries, rather than the precision or type of exposure method used. In general, maximum concentration of PM_{2.5} was higher in the studies from the USA, reaching values of up to 803 µg/m³ of PM_{2.5} in San Diego (California) (Hutchinson et al., 2018), while in Australia the maximum was of just under 295 µg/m³ of PM_{2.5} (Haikerwal et al., 2016). Additionally, other demographic and social aspects, such as population distribution within the landscape or emergency warning systems could help understand some of these differences.

Our analysis showed that short-term exposure to LFS PM_{2.5} was positively associated with asthma-related hospital admissions for up to at least 3 days after exposure and with asthma-related ED visits on the same day of exposure. Although, while we don't know the reason behind these differences, there are many factors that could contribute to the pattern of lagged associations observed in these studies. These can relate to time frames of the underlying mechanism of adverse health impacts, of the responses of people to onset of the illness, and the time taken to get healthcare and be assessed as requiring admission to hospital. While emergency department visits may increase very quickly as PM_{2.5} increases, admission to hospital may have a more delayed association because of the time taken to get assessed and treated in an emergency department, before the need for admission overnight is required. Regardless of the explanatory factors, similar results were obtained by Zheng et al. (2015) for a typical PM_{2.5} mixture. They found that there was a positive association for the short-term exposure to multi-source PM_{2.5} with the combined effect of asthma-related hospital admissions and ED visits for a lag less than or equal to 2 days and more than 2 days. Similarly, Lim et al. (2016) found a positive association for the combined effect for lags 0, 1, and 3. These results show that single day effects likely underestimate the overall impact of a 1-day exposure to LFS PM_{2.5}. However, Orellano et al. (2017) found that the lag had no influence in the association between a typical mixture PM_{2.5} and severe

asthma exacerbations.

Our descriptive analysis regarding salbutamol dispensations and medication use should be considered with caution. Although results do show a positive association between LFS $PM_{2.5}$ and salbutamol dispensations, these effects can't be completely attributed to asthma. Salbutamol dispensations were obtained from a pharmaceutical database administered by British Columbia, which records every prescription dispensed. Salbutamol may be used to relieve exacerbations of asthma, chronic obstructive pulmonary disease (COPD) and other obstructive lung diseases (Elliott et al., 2013; Yao et al., 2013). Additionally, although we know that prescriptions increased, we are not sure on the magnitude of increase in the actual use of these. In the case of Johnston et al. (2006), medication use was identified and quantified through a self-reported daily diary, which means that there is a high risk of bias in the outcome measures.

Overall, we found larger positive associations for hospital admissions and ED visits compared to multisource air pollution. During short fire periods, people are exposed to higher concentrations of $PM_{2.5}$, therefore our results suggest that the shape of the exposure response function might be steeper at higher $PM_{2.5}$ concentrations. Additionally, fire episodes happen during warm seasons (usually summer), and these findings were similar in direction to those obtained by Lim et al. (2016), Fan et al. (2016), and Zheng et al. (2015), where effects were considerably higher for warm season compared to cold. A recent study examining the cardiopulmonary effects of $PM_{2.5}$ exposure among older adults obtained similar results, with a higher percentage increase of asthma-related (asthma, bronchitis or wheezing) hospitalizations during smoke days [6.9% (95% CI: 3.71, 10.11)] compared to non-smoke days [1.34% (95% CI: -1.1, 3.77)] (DeFlorio-Barker et al., 2019). Results for smoke days are similar in magnitude and direction to those obtained in our meta-analysis. Likewise, Elliott et al. (2013) estimated that LFS $PM_{2.5}$ was positively associated with salbutamol dispensation with a higher effect during extreme fire days (RR = 1.07, 95% CI: 1.04–1.09) compared to fire season average (RR = 1.06, 95% CI: 1.04–1.07), for a $10 \mu g/m^3$ of $PM_{2.5}$. However, we don't see these type of differences for cardiovascular outcomes (Cascio, 2018), so magnitude might not be the only explanation. Somehow, respiratory outcomes and particularly asthma, behave differently and this might be explained by the presence of other pollutants, such as volatile organic compounds (VOCs) which are emitted during wildfires and may cause coughing and wheezing (Yousouf et al., 2014).

To the best of our knowledge, this is the first meta-analysis to obtain summary estimates between LFS $PM_{2.5}$ and asthma-related outcomes. Our results suggest that the magnitude of the effects estimate for LFS tend to be larger than those from general multi-source $PM_{2.5}$ air pollution, and population sub-groups are affected in a different manner, elders being more vulnerable than adults, and females more susceptible than males.

There are several limitations to this study. First, the number of studies included in each meta-analysis was small and heterogeneity was high. The studies assessed also have high variability in the statistical methods applied, covariates adjusted for, exposure measurement and modelling, and important geographical and temporal differences. Additionally, we combined different types of effects estimates (OR, RR, % increase) by transforming them to a common risk ratio per 10-unit increase in $PM_{2.5}$, although these should not be an issue as the incidence of asthma-related outcomes particularly during fire episodes could be considered a rare outcome. Second, most studies were done in Australia, the USA, or Canada, while analysis of other fire-prone areas are limited with respect to asthma-related outcomes. In part, the consistency of results seen within this analysis may be attributed to the fact that there is a low heterogeneity with respect to location and populations analysed within the selected studies, and probably because the number of research groups addressing LFS health impacts is also limited. Nevertheless, we did observe a wide variety of methods used within these analyses, including simple to sophisticated exposure

modelling methods and comparison of different methods within same studies. Further, if two studies covered the same population and time period only one was included for the quantitative analysis. It is possible that these results might not be generalised to developing countries or economies in transition, and therefore we highlight the need for new studies, particularly in low- and middle-income countries (LMIC). Third, there could be a potential bias related to how health systems classifies health outcomes in each of the countries analysed, where outcomes such as ED visits, hospital admissions, physician visits, and outpatient visits are not defined and treated equivalently.

Our sensitivity analyses aimed to produce results as robust as possible in the context of existing limitations. These analyses showed that the direction of association was maintained independent of which studies were included.

5. Conclusion

Our systematic review and meta-analysis suggests that asthma-related hospital admissions and emergency department visits are positively associated with LFS $PM_{2.5}$, and some groups such as women, adults, and elders are more affected than others. Hospital admissions results show that smoke effects could last for multiple days, while for ED visits effects tended to occur on the same day as exposure. Currently there are no recommended coefficients for asthma-related outcomes, apart from asthma incidence in children, when undertaking health impact assessments. Usually, these recommended values tend to focus on mortality and cardiovascular and respiratory morbidity (WHO, 2013). In recent years, at least three meta-analyses (Fan et al., 2016; Lim et al., 2016; Zheng et al., 2015) have estimated pooled effects for short-term exposure to $PM_{2.5}$ and asthma-related hospital admissions and/or emergency department visits, but none have focused on LFS $PM_{2.5}$.

Given the expected increase in longer and more extreme fire events, there is a need to adequately assess strategies that will help reduce public health impacts. The results reported from this meta-analysis suggest that by using previously existing estimates or non-fire specific estimates, the total public health impacts and costs are likely to be underestimated.

Authors contributions

NB, FJ, AP and GM participated in the design of the study and protocol. NB and FJ participated in the coordination. NB drafted the manuscript. NB and JH performed the study screening, selection and data extraction. NB and RT did the quality assessment. NB, FJ and GM defined the overall focus of the study. NB performed the statistical analysis. All authors critically reviewed the manuscript for its intellectual content, and approved the final manuscript.

Declaration of competing interest

The authors declare they have no actual or potential conflicts of interest.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.envres.2019.108777>.

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